Practical 7

Develop a program to manage resource allocation for five processes (Google Drive, Firefox, Word Processor, Excel, and PowerPoint) using four types of resources (Printer, ROM, Hard Disk, and RAM). The program takes input for allocated, maximum, and available resources, calculates the current need of each process, and determines if a safe execution order exists using the Banker's Algorithm.

```
  GNU nano 8.7                                                    banker.c
#include <stdio.h>

#define P 5    // Number of Processes
#define R 4    // Number of Resources

int main()
{
    int allocation[P][R], max[P][R], need[P][R];
    int available[R];
    int finish[P] = {0};
    int safeSeq[P];
    int work[R];
    int i, j, k, count = 0;

    printf("\nEnter Allocation Matrix (5x4):\n");
    for(i = 0; i < P; i++)
    {
        for(j = 0; j < R; j++)
        {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Maximum Matrix (5x4):\n");
    for(i = 0; i < P; i++)
    {
        for(j = 0; j < R; j++)
        {
```

```
        }
    }

    // Banker's Algorithm
    while(count < P)
    {
        int found = 0;

        for(i = 0; i < P; i++)
        {
            if(finish[i] == 0)
            {
                for(j = 0; j < R; j++)
                {
                    if(need[i][j] > work[j])
                        break;
                }

                if(j == R)
                {
                    for(k = 0; k < R; k++)
                        work[k] += allocation[i][k];

                    safeSeq[count] = i;
^G Help        ^O Write Out   ^F Where Is    ^K Cut
^X Exit        ^R Read File   ^\ Replace     ^U Paste
```

```
            printf("\nSystem is NOT in Safe State (Deadloc
            return 0;
        }
    }

    printf("\nSystem is in SAFE State\nSafe Sequence:\n");

    char *processes[] = {
        "Google Drive",
        "Firefox",
        "Word Processor",
        "Excel",
        "PowerPoint"
    };

    for(i = 0; i < P; i++)
    {
        printf("%s", processes[safeSeq[i]]);
        if(i != P-1)
            printf(" -> ");
    }

    printf("\n");

    return 0;
}
```

```
piyus@piyush MINGW64 ~
$ mkdir -p /c/sys64/home/piyus@piyush/Practica07

piyus@piyush MINGW64 ~
$ nano banker.c

piyus@piyush MINGW64 ~
$ gcc banker.c -o banker

piyus@piyush MINGW64 ~
$ ./banker

Need Matrix:
0 6 4 2
0 0 2 0
0 0 0 0
0 7 5 0
1 0 0 2

System is in SAFE state.
Safe Sequence:
Firefox -> WordProcessor -> Excel -> PowerPoint -> GoogleDrive -> END

piyus@piyush MINGW64 ~
$
```